Measurement of Absolute Double Differential Cross-Section in Invariant Mass and p_T Bins

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1 Input Files and Bin Ranges

Input Data Files:

- Merged ROOT files containing reconstruction and hodoscope efficiencies.
- **Roadsets:** RS57, RS59, RS62, RS67 (run 3089), and RS70.
- **Targets:** Liquid Hydrogen (LH2), Liquid Deuterium (LD2), and Empty Flask.

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- p_T **Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- x_F **Bins:** [0.0, 0.85] with a fixed step width of 0.05.

2 Event Selection Criteria

Dimuon Kinematics:

- $4.2 < Mass < 8.8$ GeV, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos\theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

Track Quality & Acceptance:

- $\chi^2_{target} < 15$, $\chi^2_{dimuon} < 18$, and $\chi^2/(nHits - 5) < 12$.
- Hit requirements: nHits > 13 per track, $nHits_1 + nHits_2 > 29$, St1 total > 8.
- Track vertex longitudinal limits: $-320 < z_v < -5$.

- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for runID ≥ 11000 and 0.4 otherwise.

Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.

3 Plots and Distributions

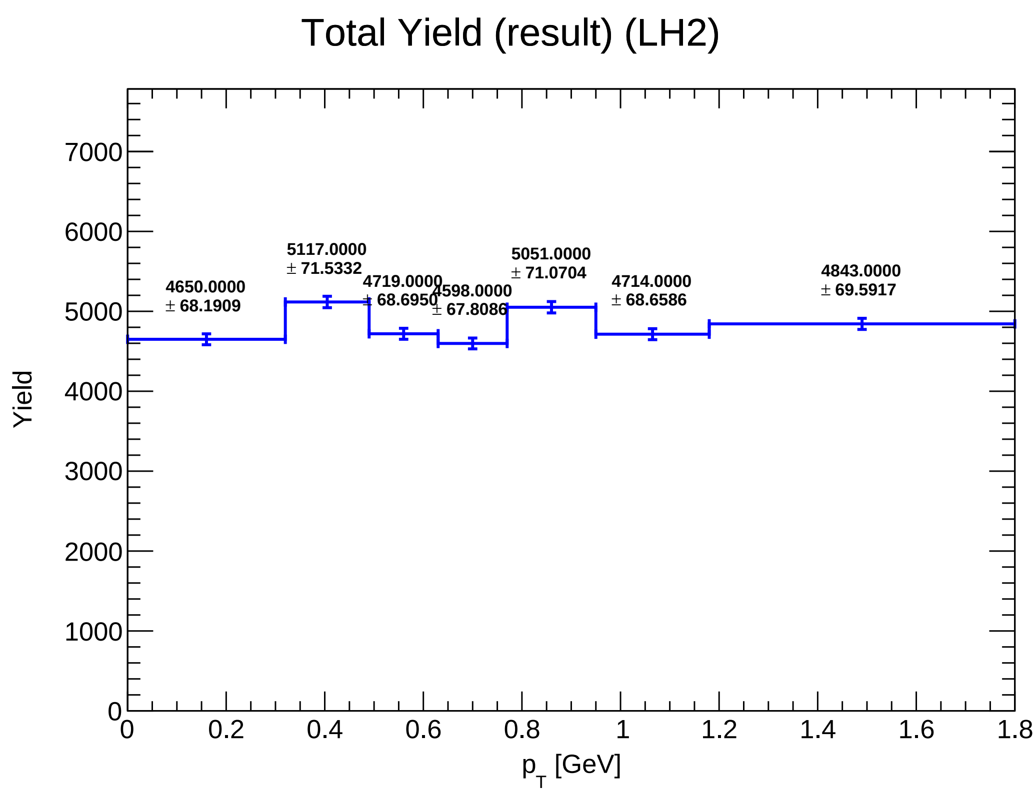

Figure 1: Y_total_LH2

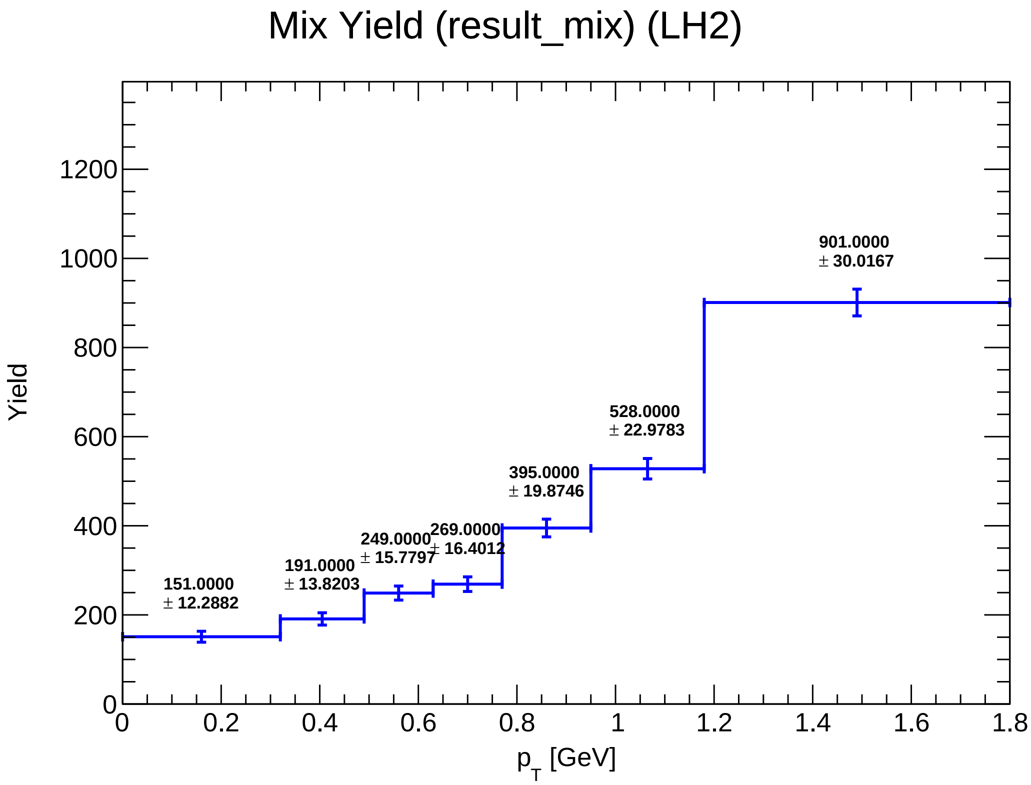

Figure 2: Y_mix_LH2

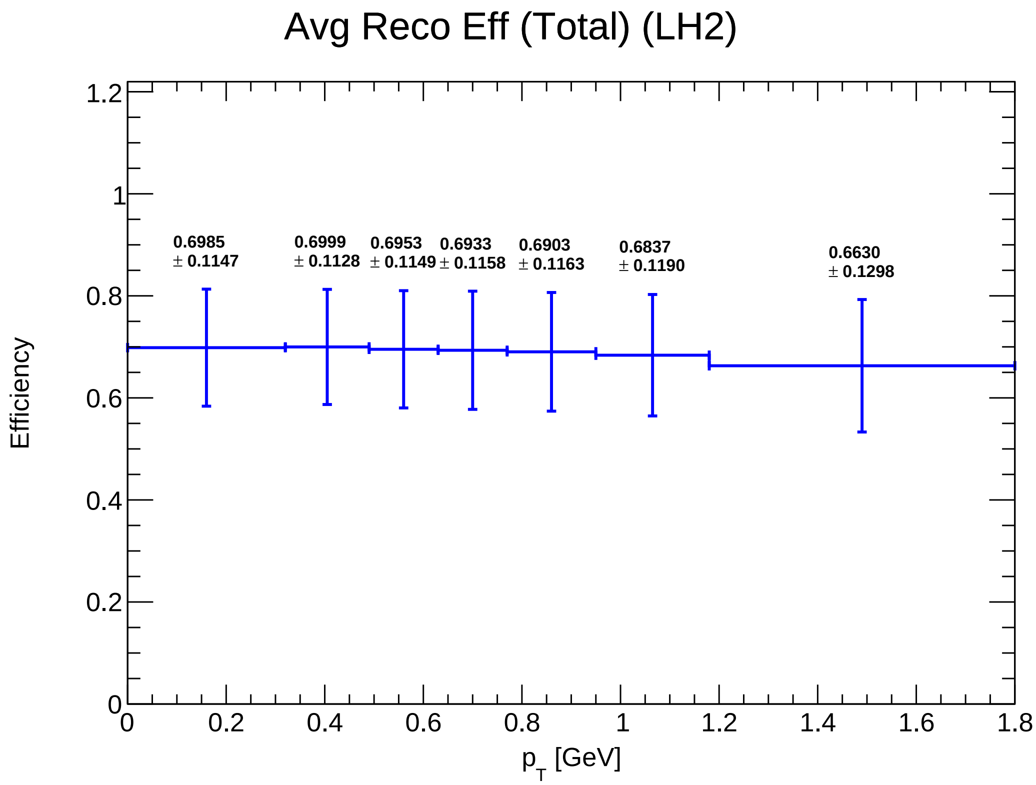

Figure 3: E_total_reco_LH2

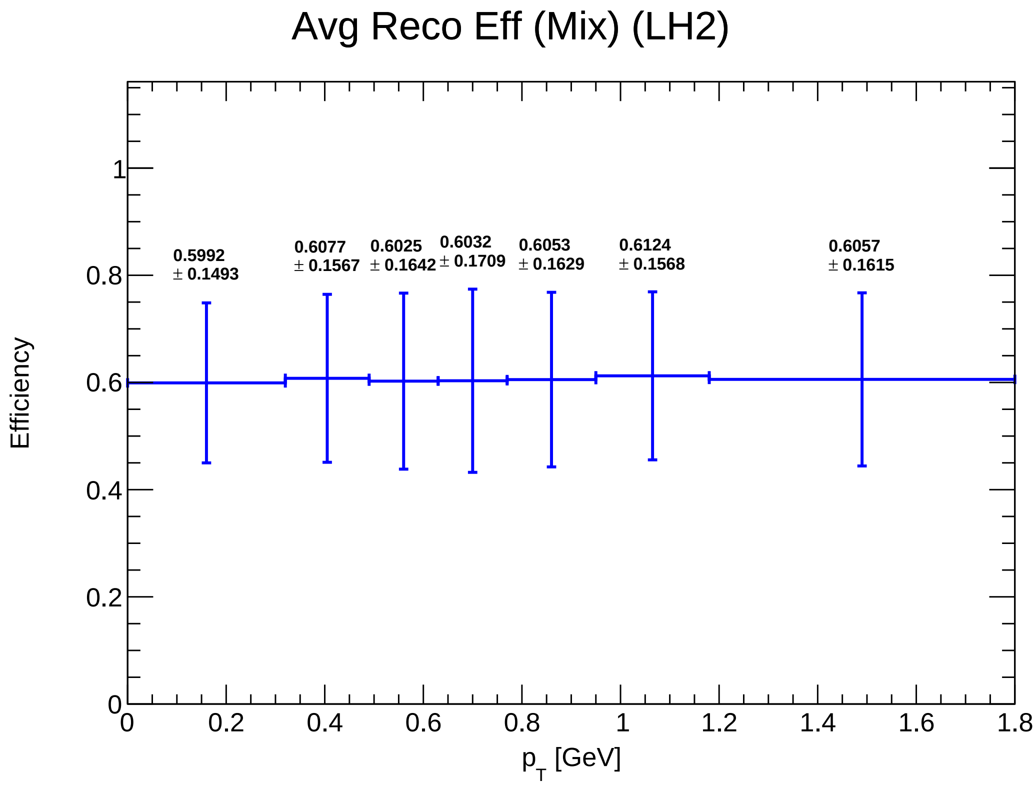

Figure 4: E_mix_reco_LH2

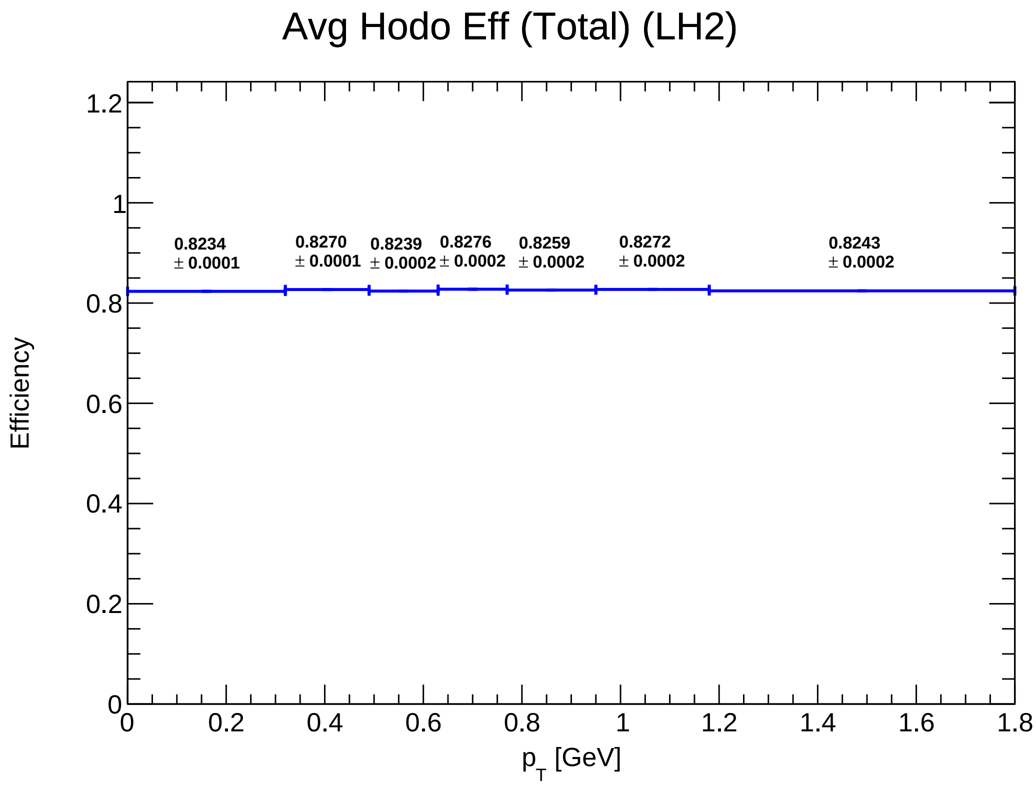

Figure 5: E_total_hodo_LH2

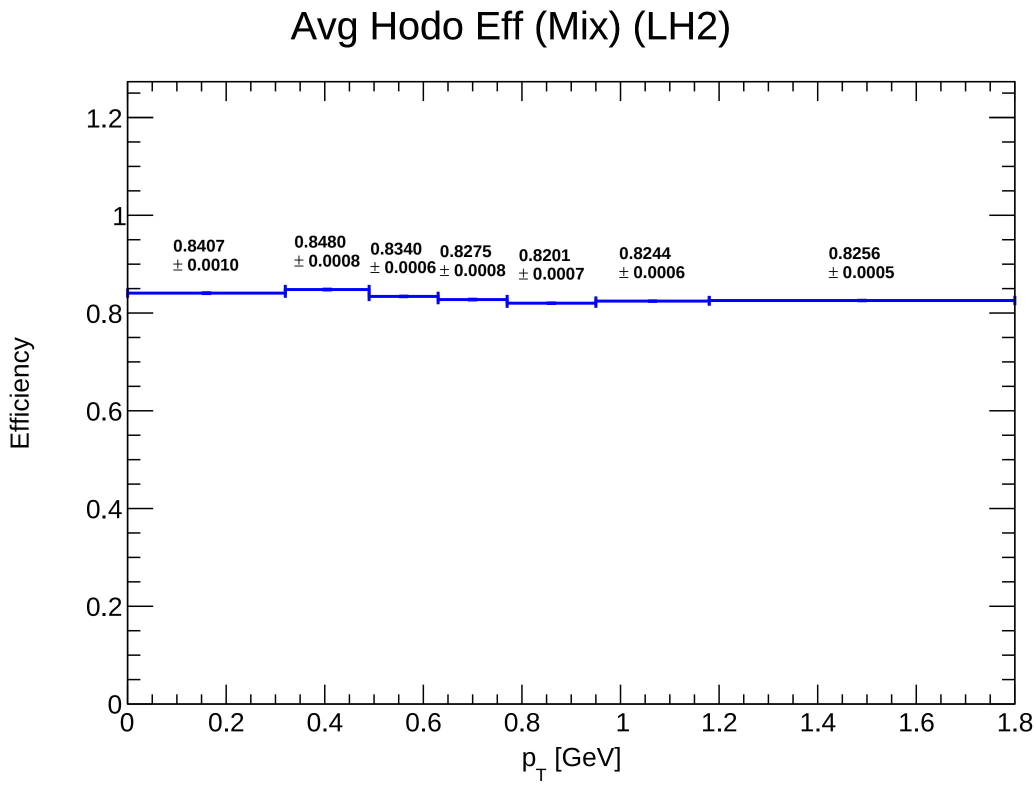

Figure 6: E_mix_hodo_LH2

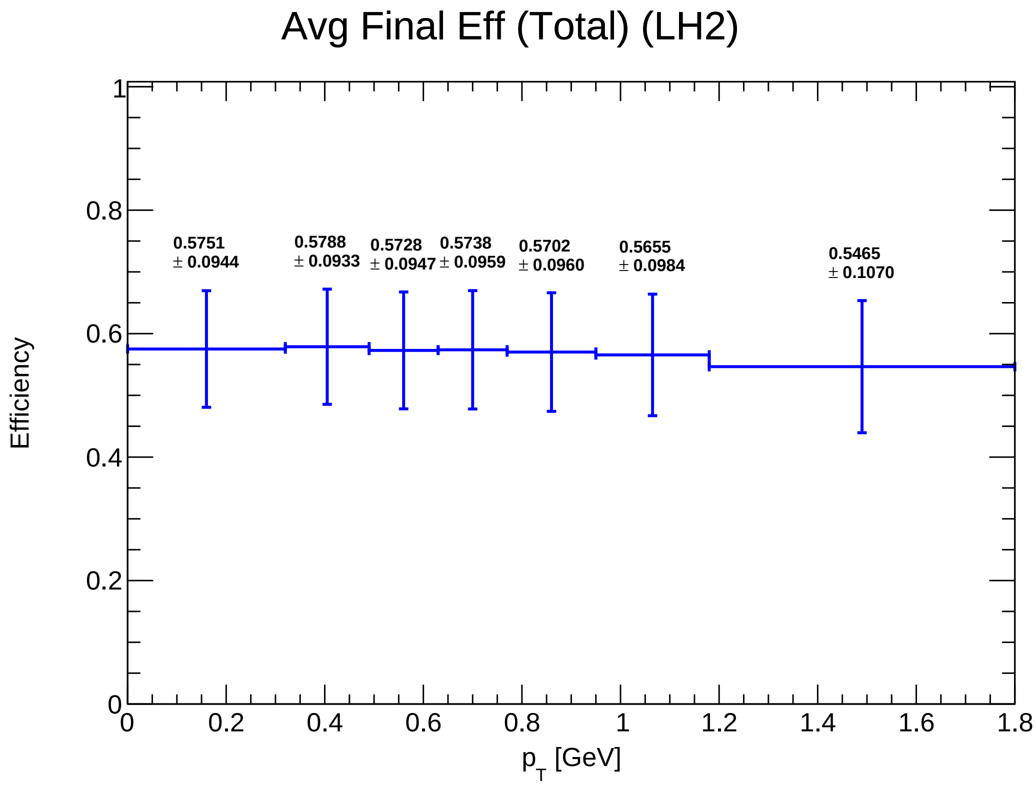

Figure 7: E_total_final_LH2

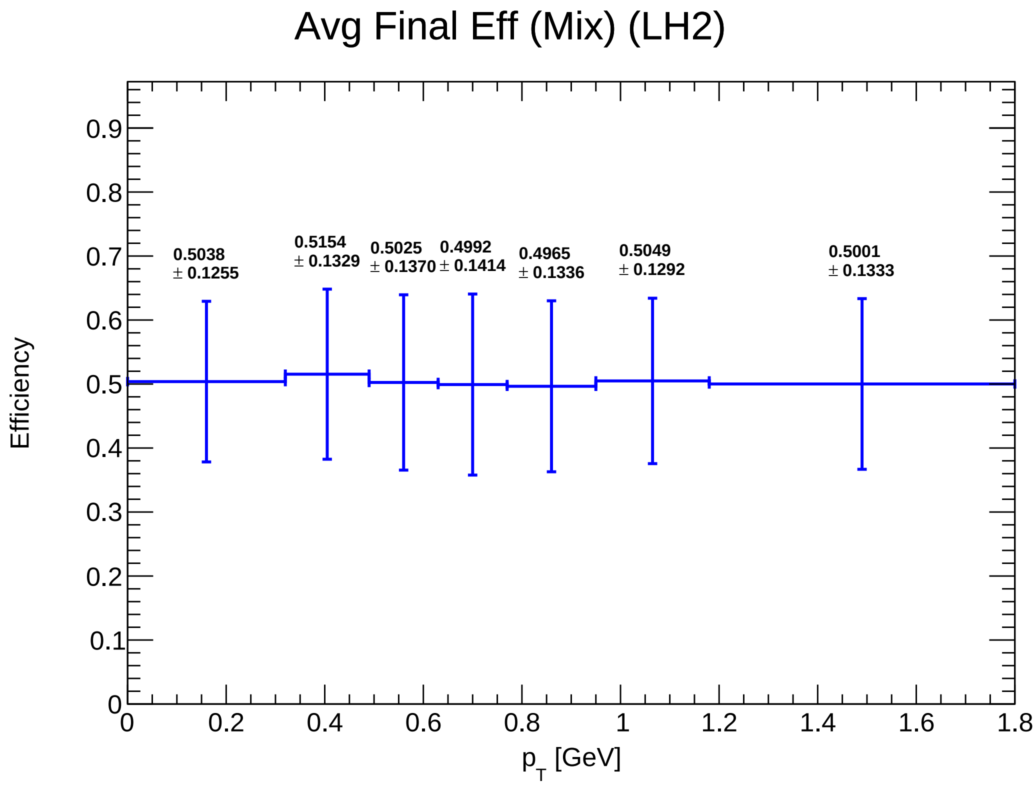

Figure 8: E_mix_final_LH2

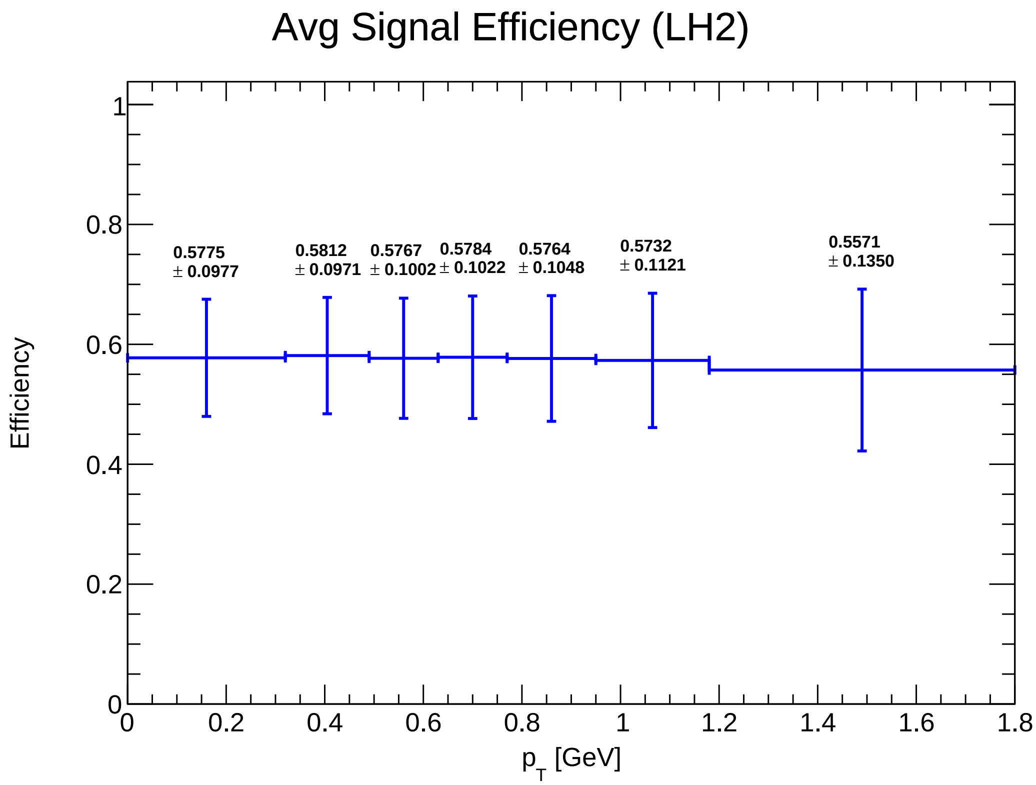

Figure 9: E_final_signal_LH2

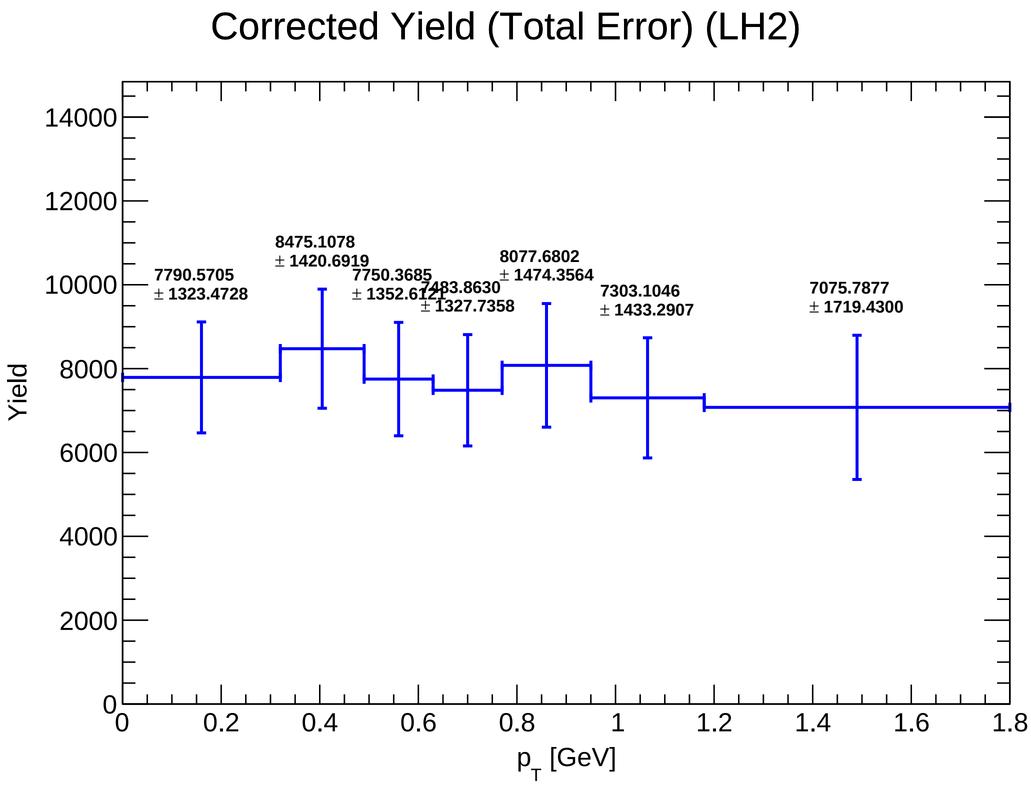

Figure 10: Y_corrected_LH2

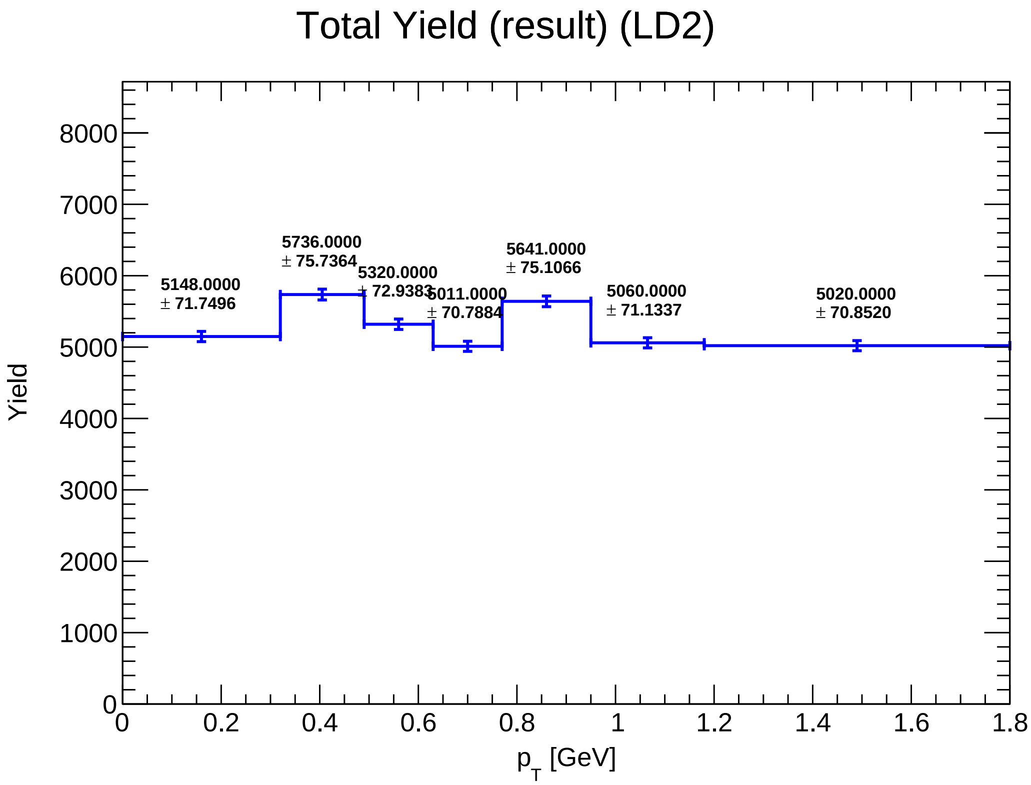

Figure 11: Y_total_LD2

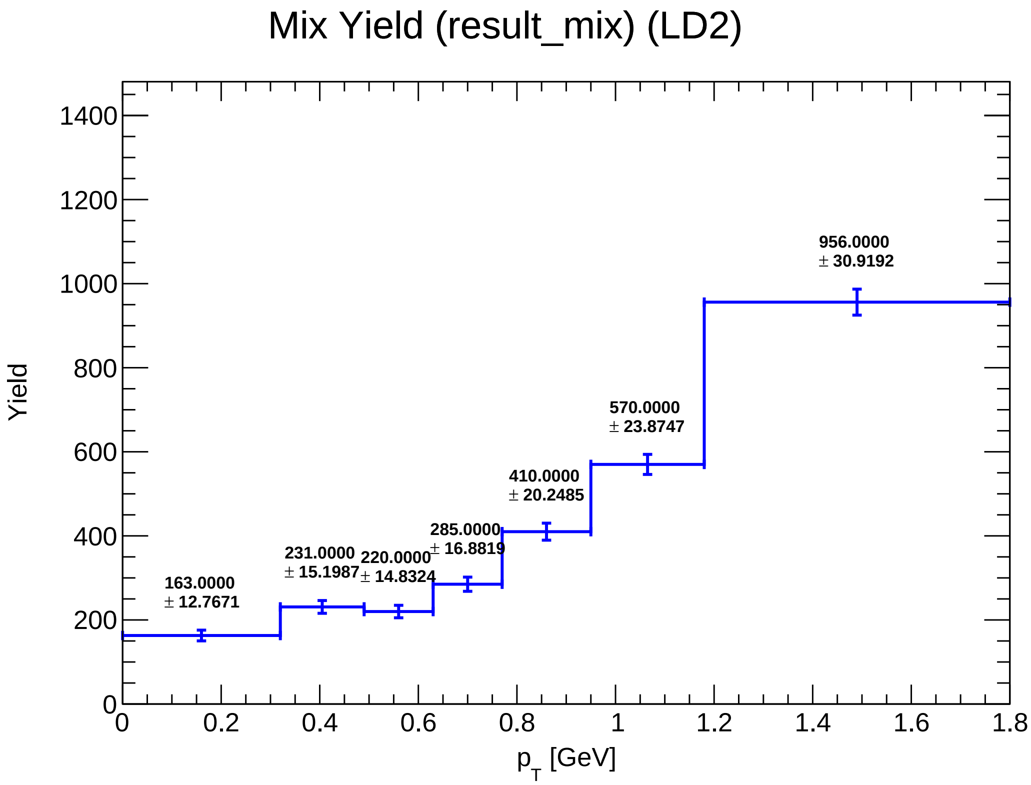

Figure 12: Y_mix_LD2

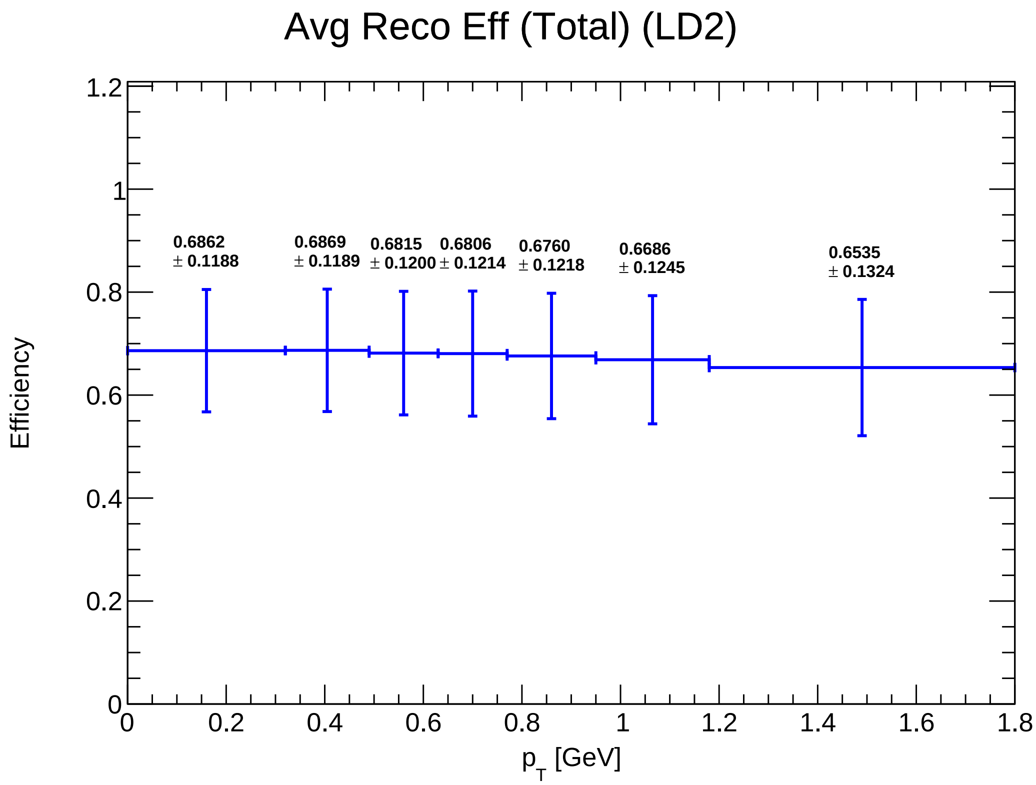

Figure 13: E_total_reco_LD2

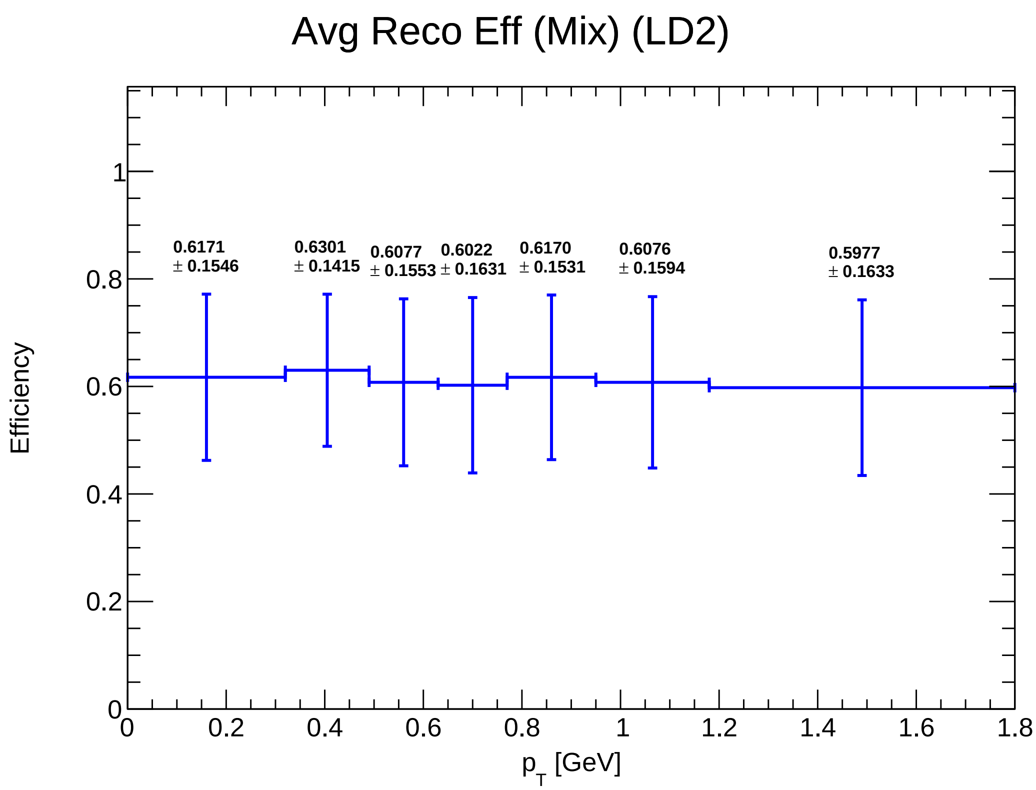

Figure 14: E_mix_reco_LD2

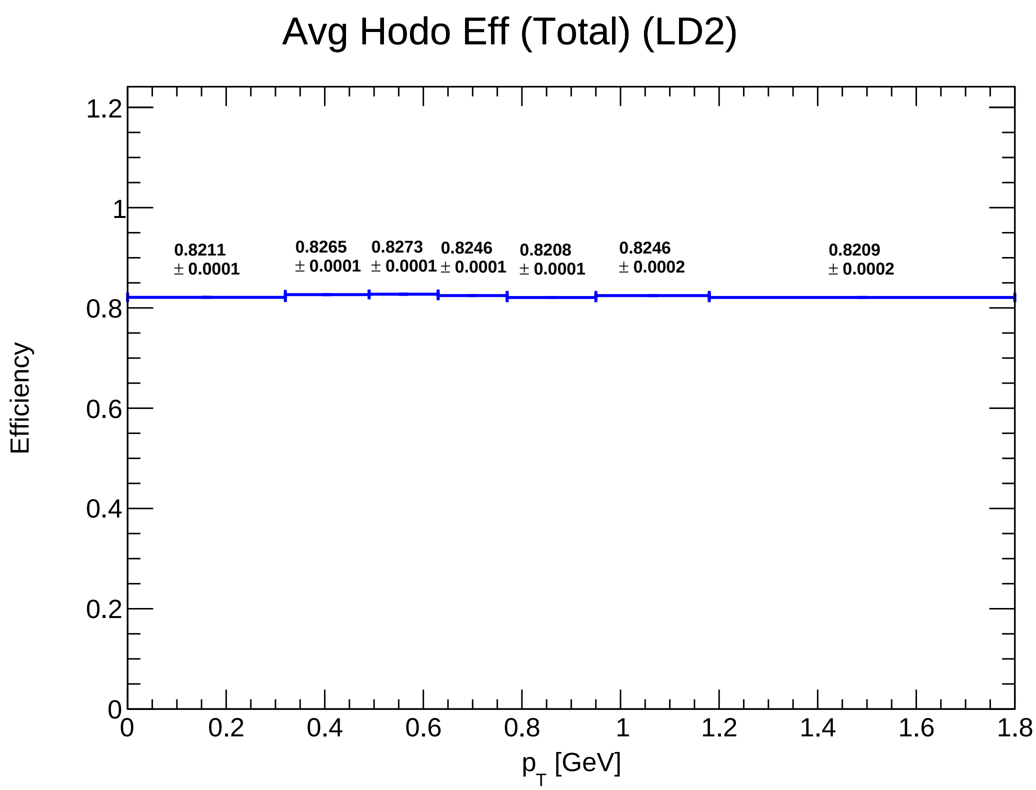

Figure 15: E_total_hodo_LD2

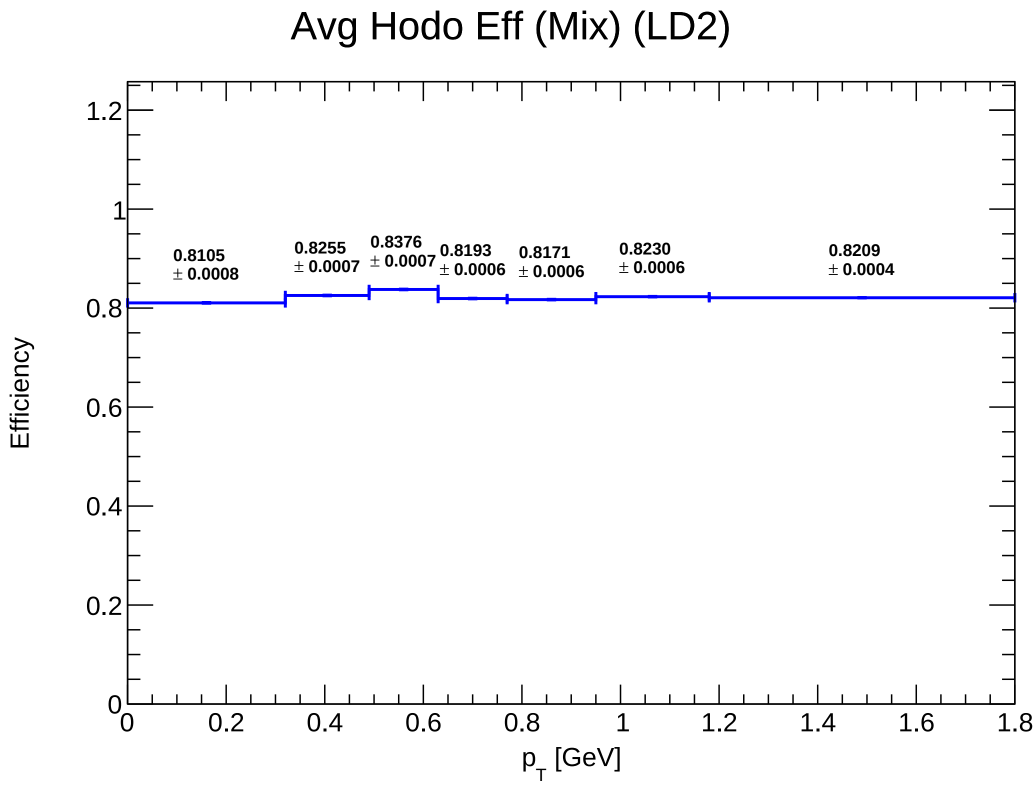

Figure 16: E_mix_hodo_LD2

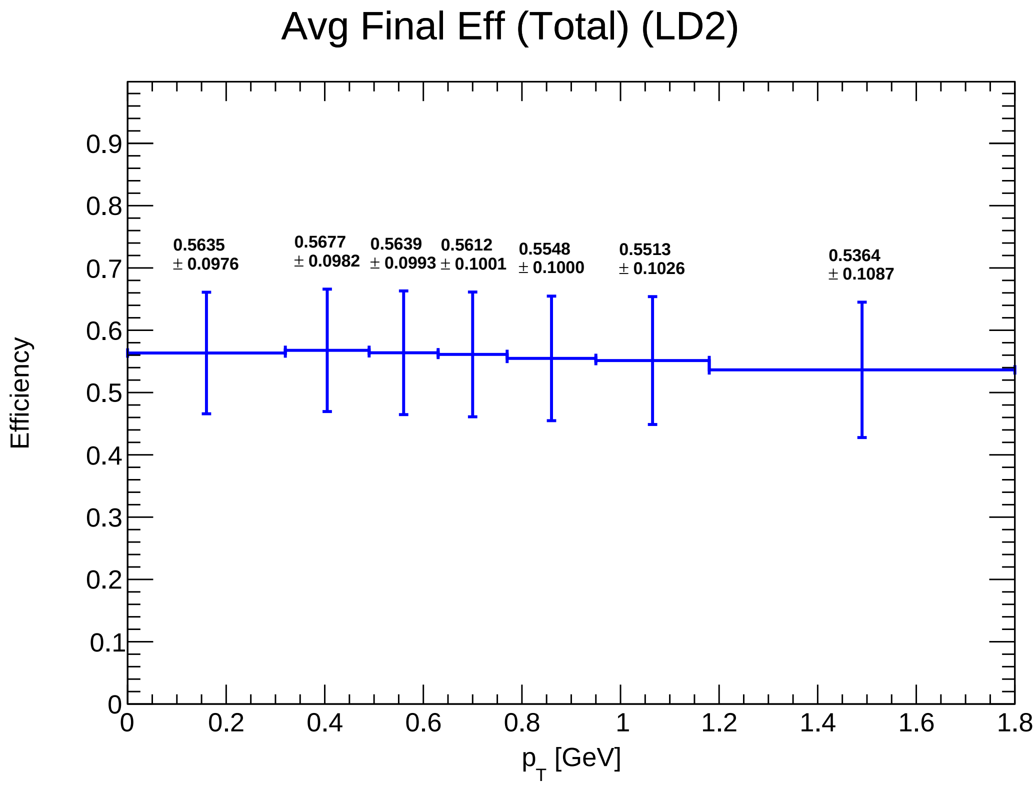

Figure 17: E_total_final_LD2

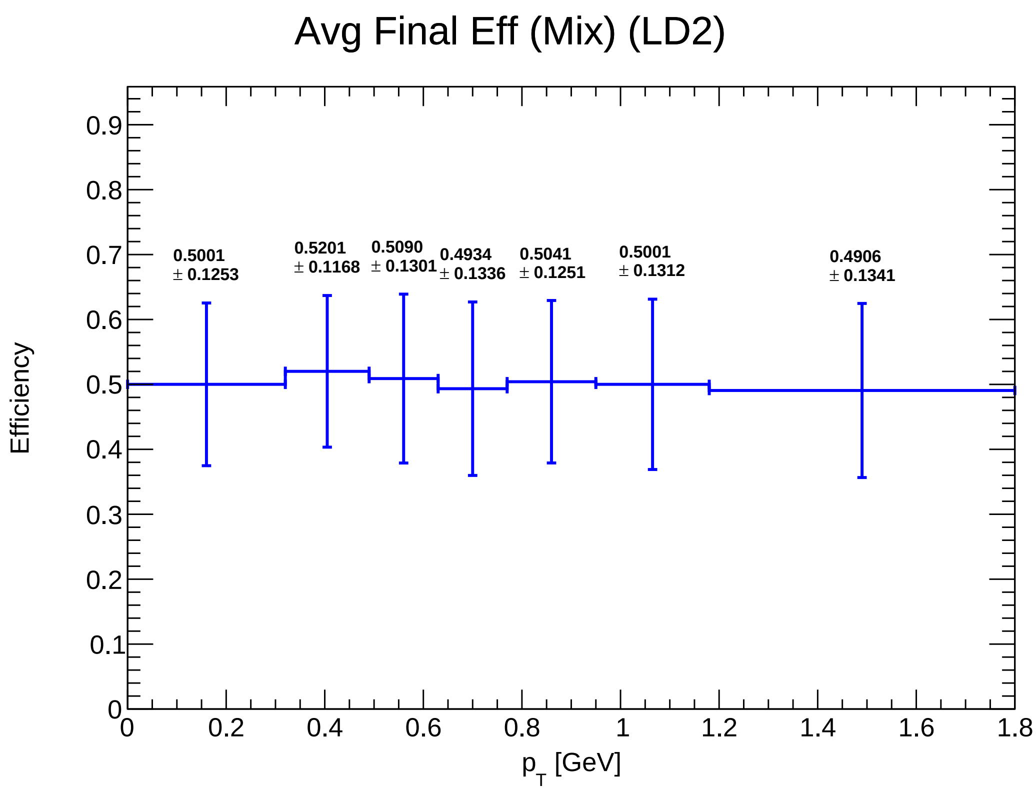

Figure 18: E_mix_final_LD2

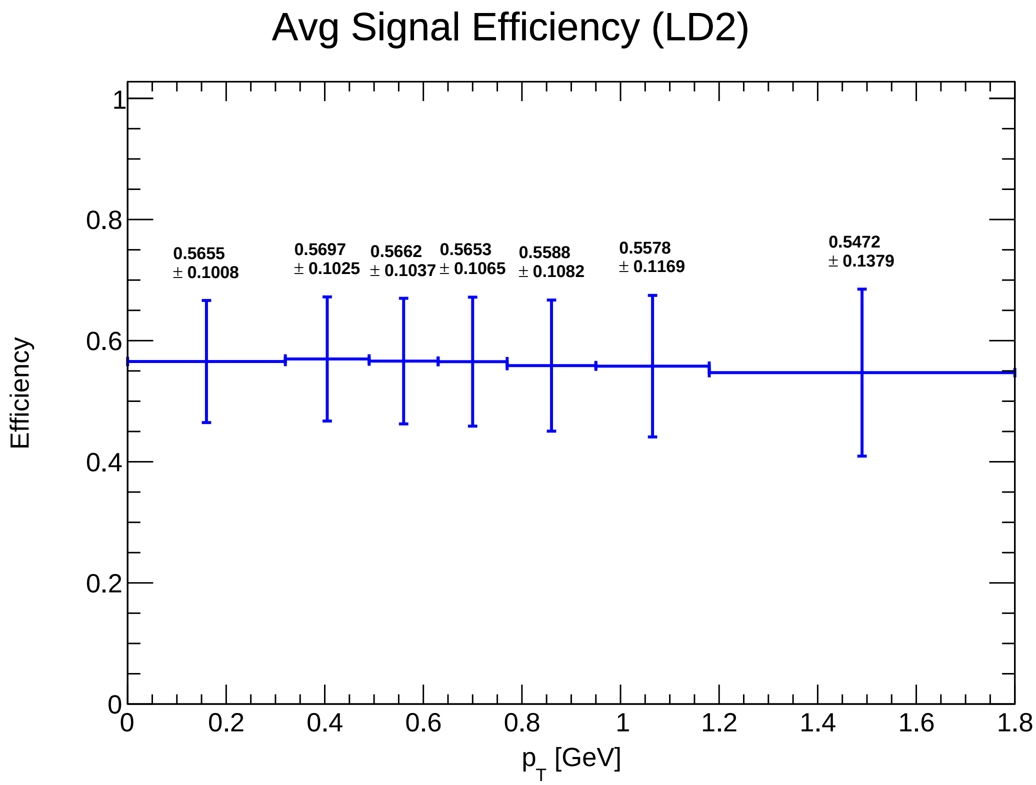

Figure 19: E_final_signal_LD2

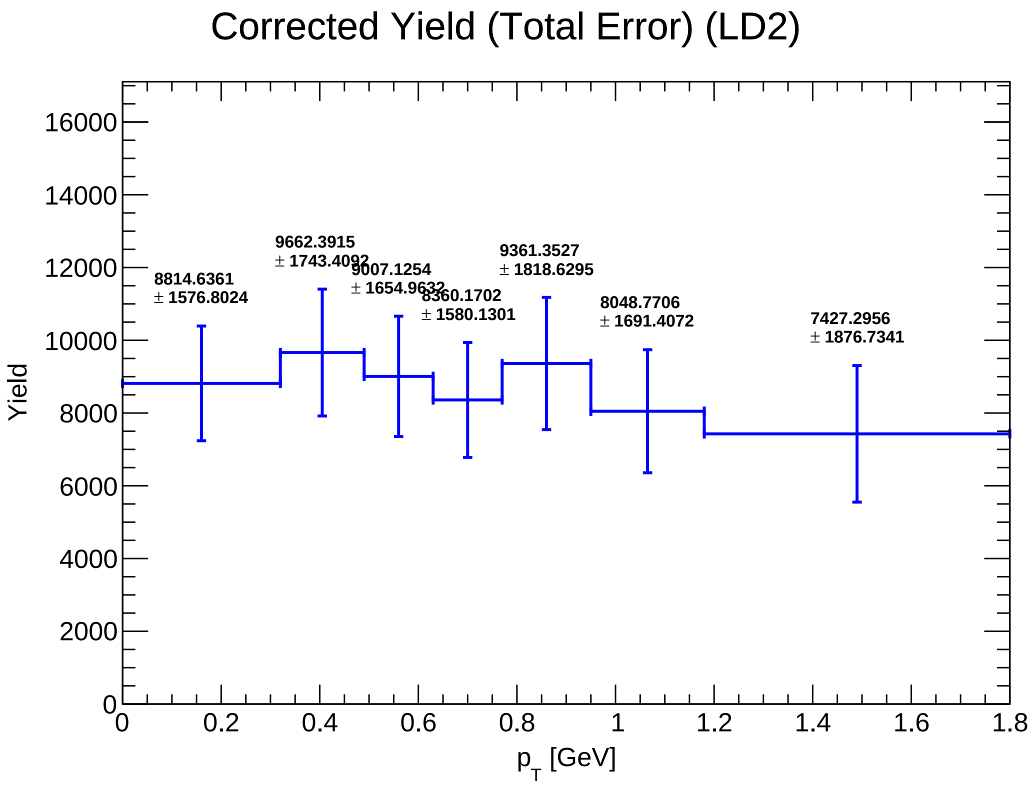

Figure 20: Y_corrected_LD2

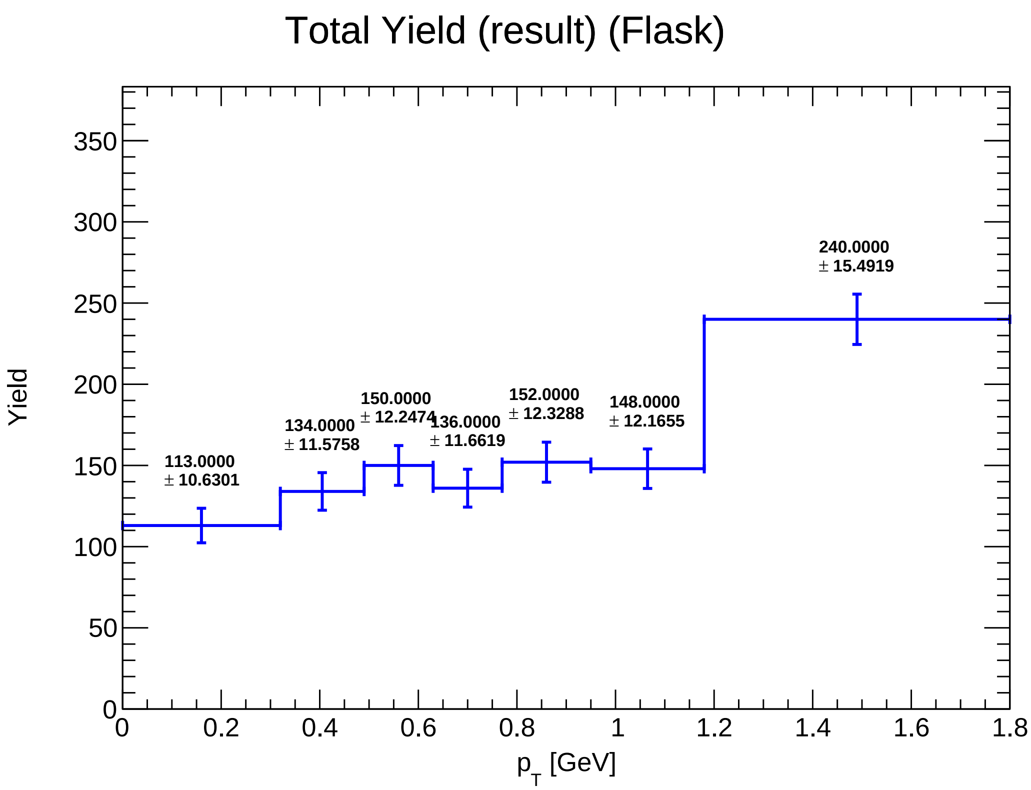

Figure 21: Y_total_Flask

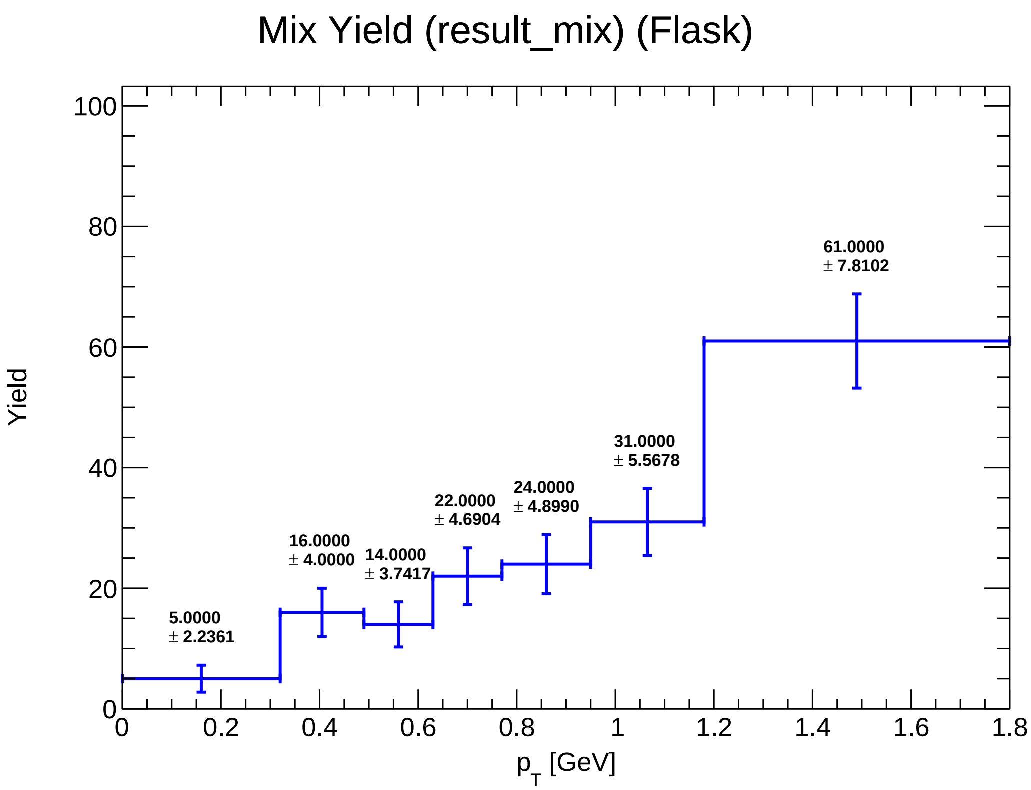

Figure 22: Y_mix_Flask

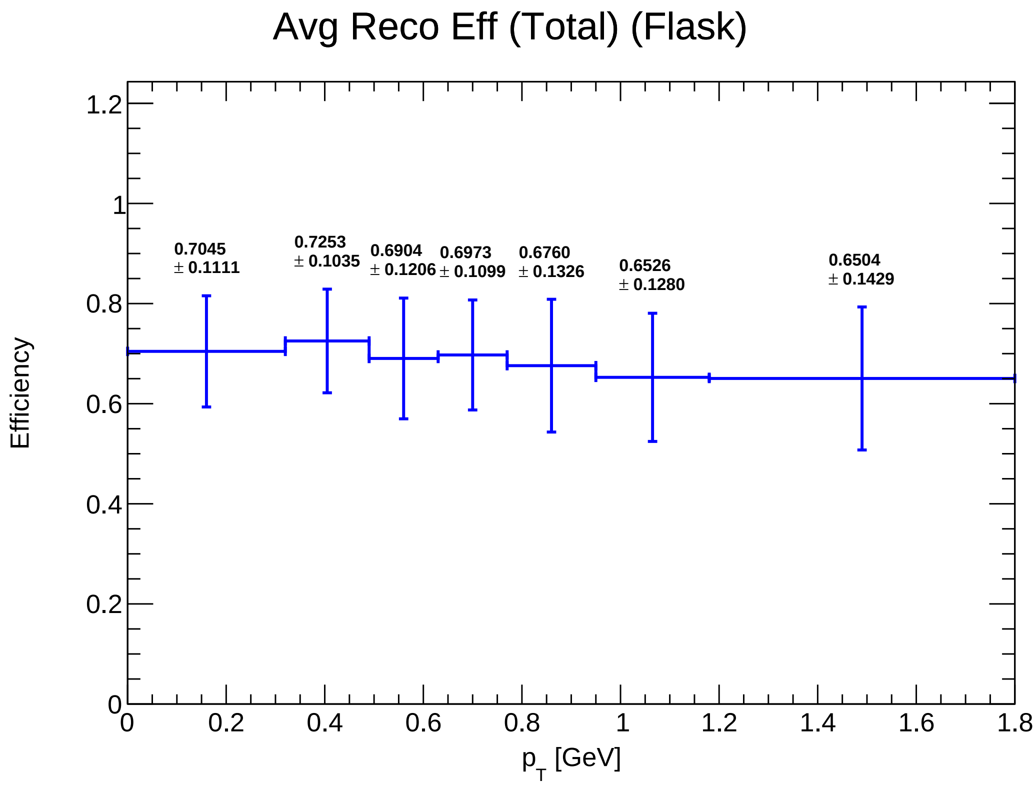

Figure 23: E_total_reco_Flask

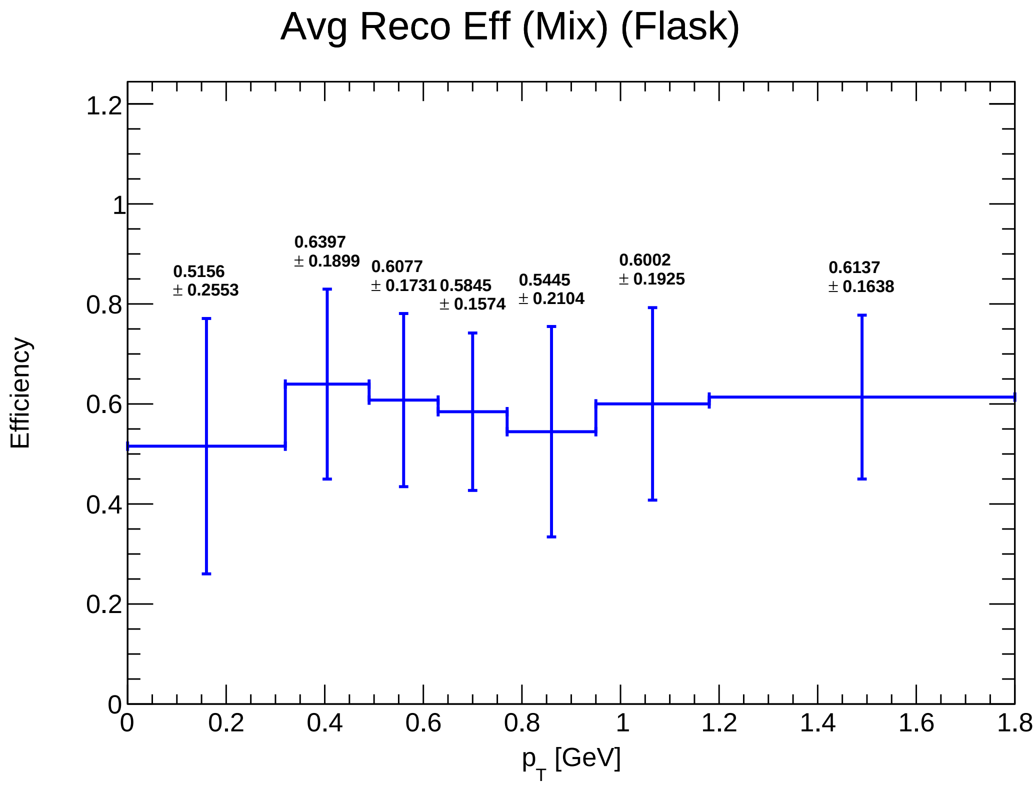

Figure 24: E_mix_reco_Flask

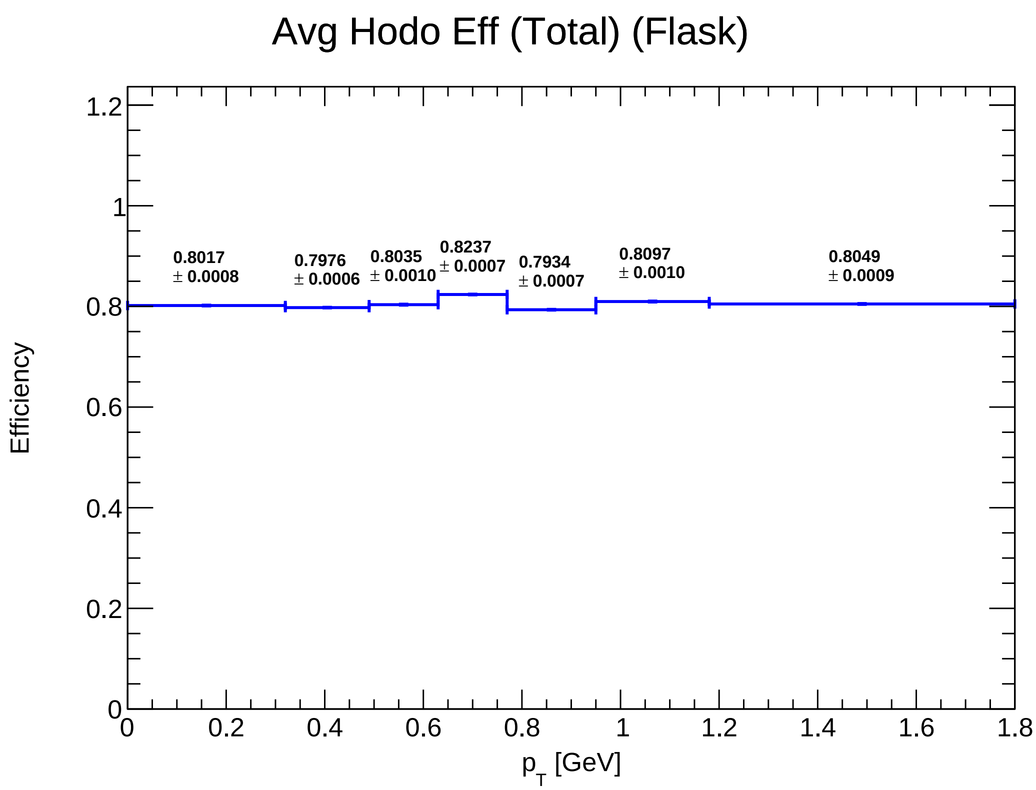

Figure 25: E_total_hodo_Flask

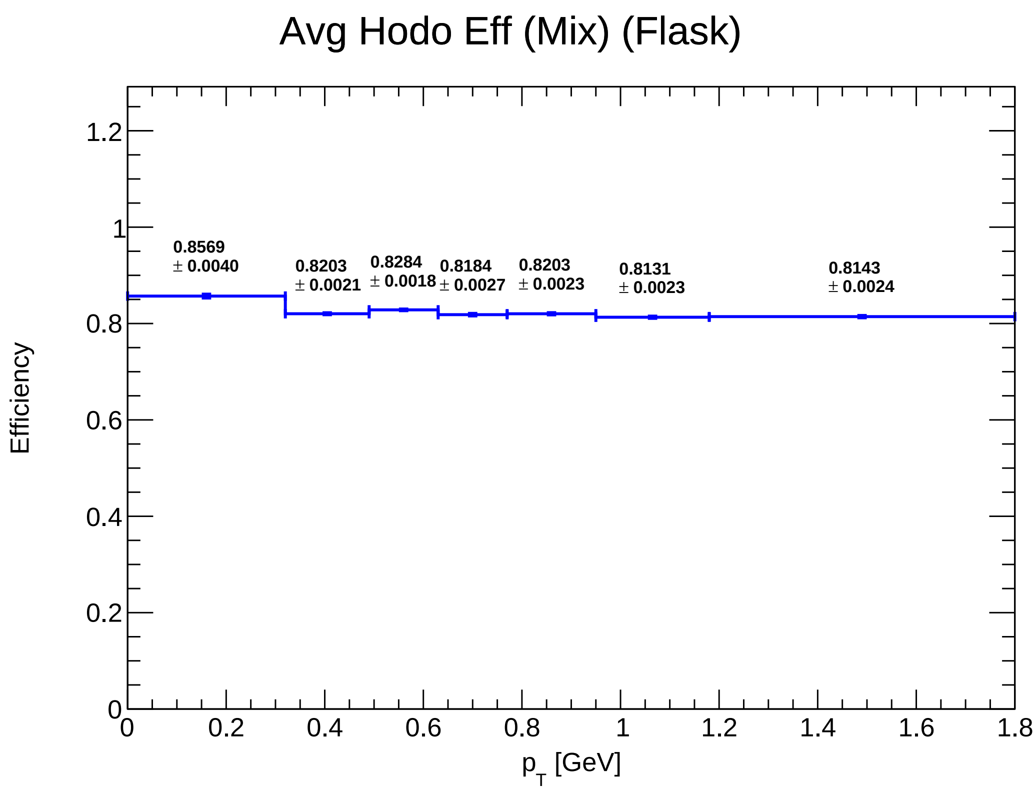

Figure 26: E_mix_hodo_Flask

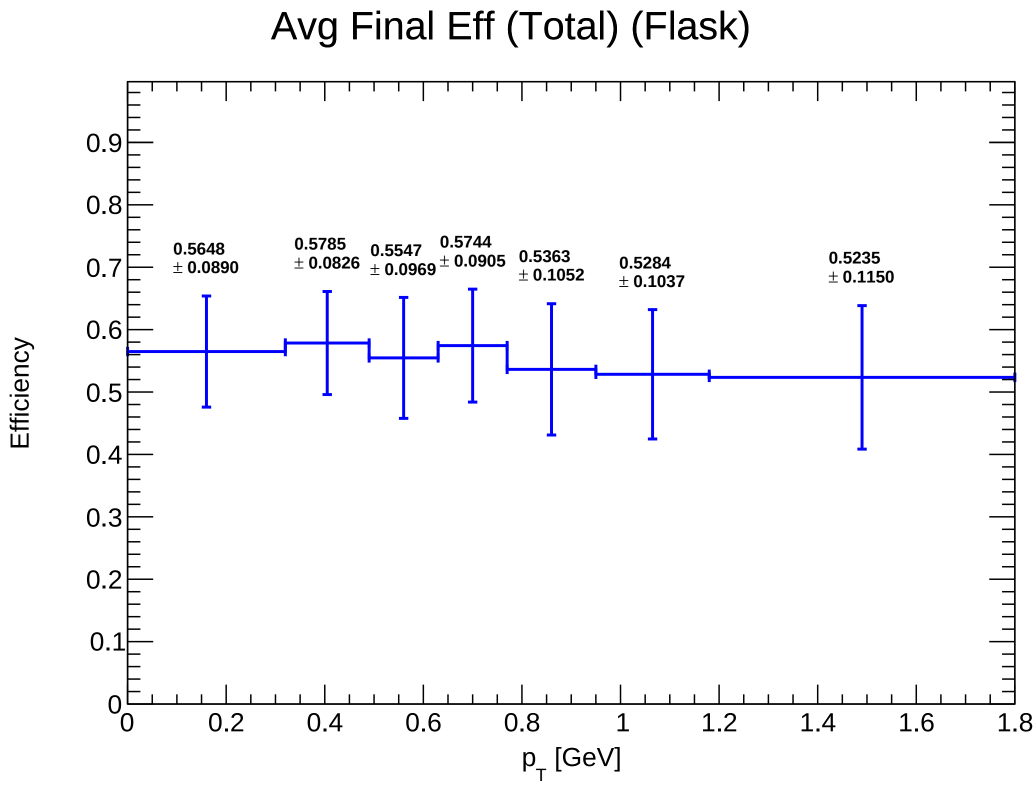

Figure 27: E_total_final_Flask

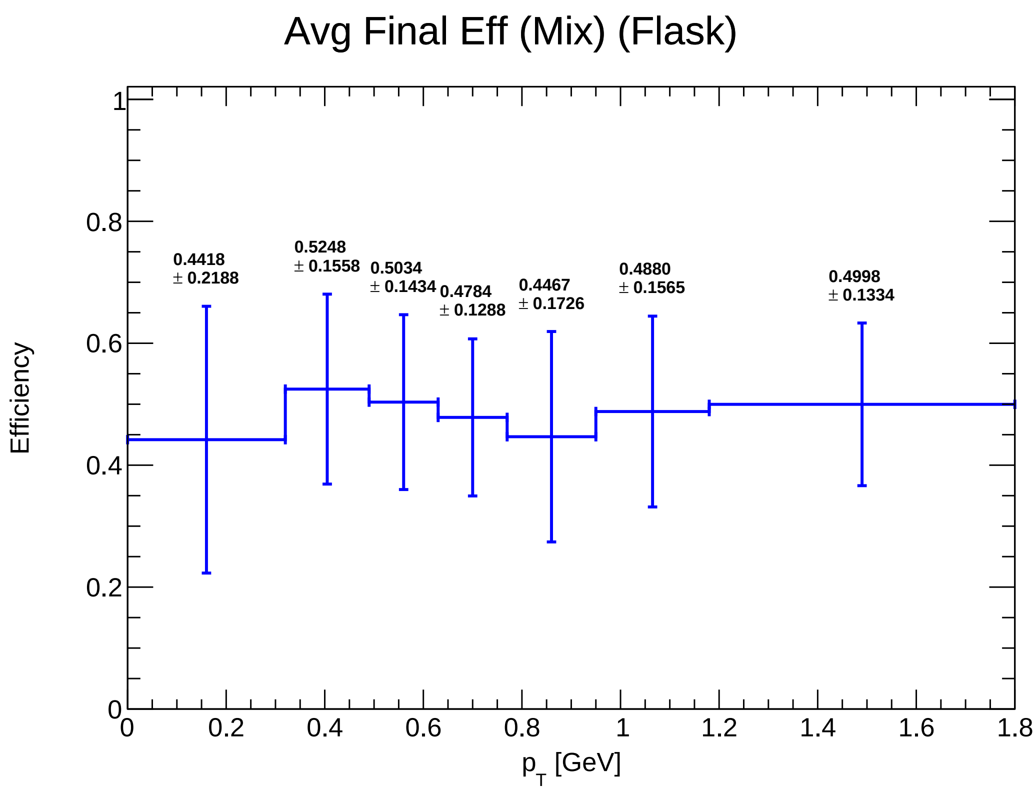

Figure 28: E_mix_final_Flask

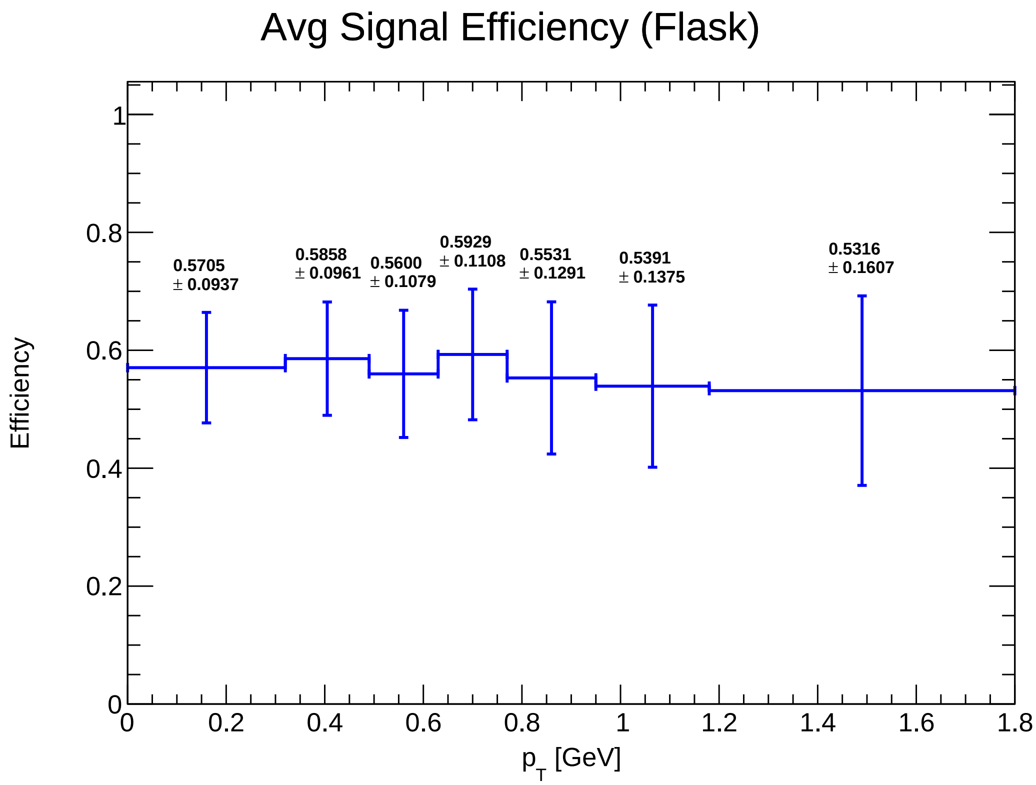

Figure 29: E_final_signal_Flask

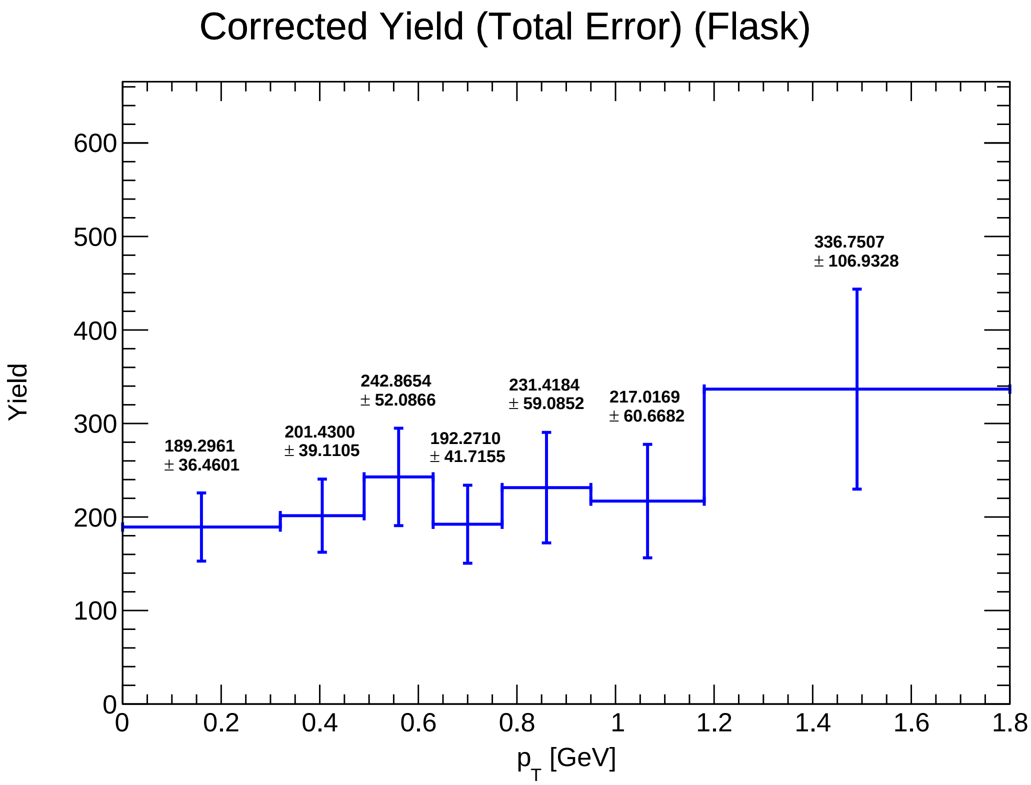

Figure 30: Y_corrected_Flask

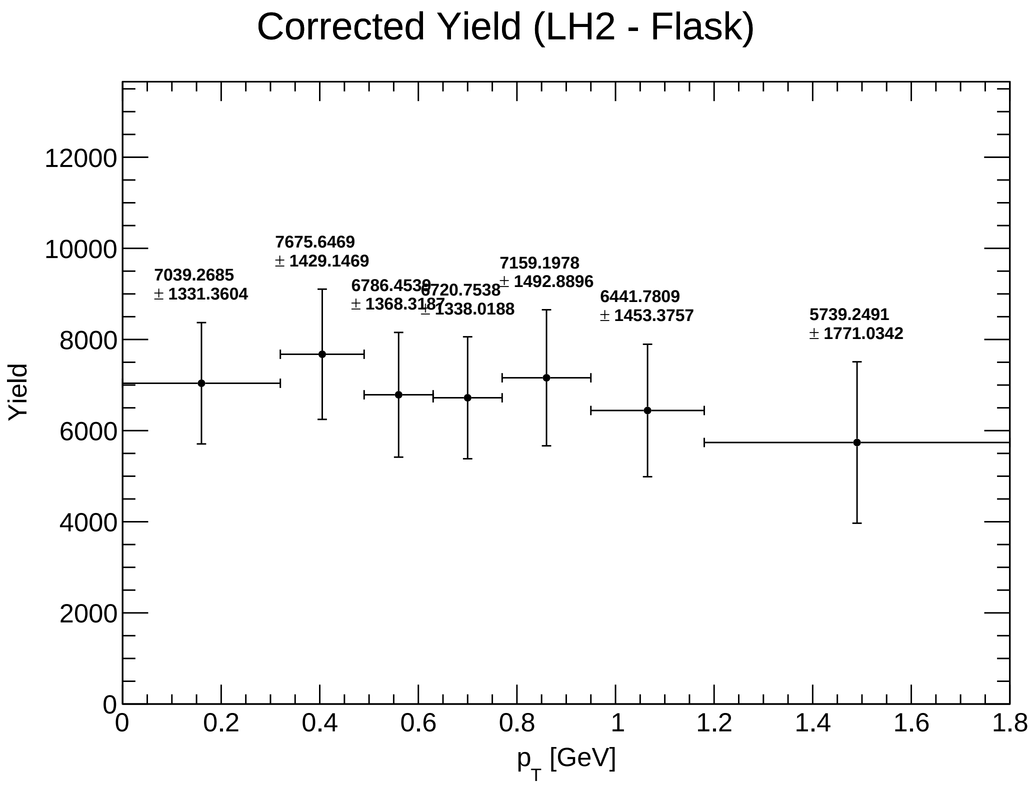

Figure 31: Y_corrected_Subtracted_LH2

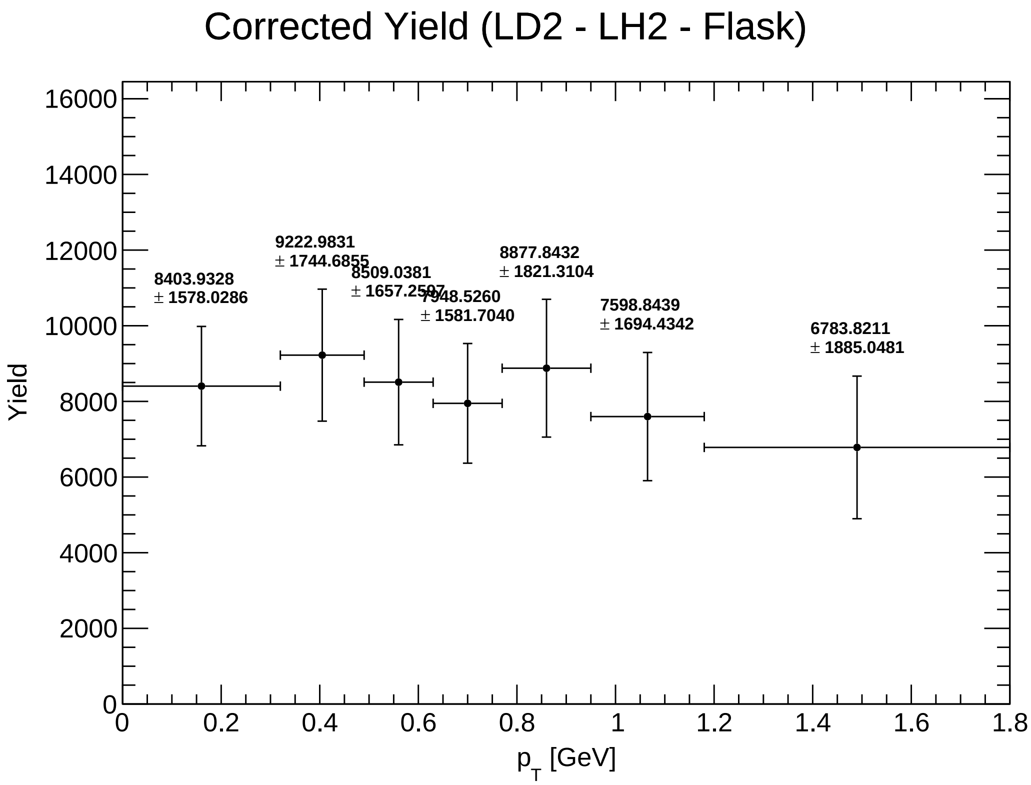

Figure 32: Y_corrected_Subtracted_LD2

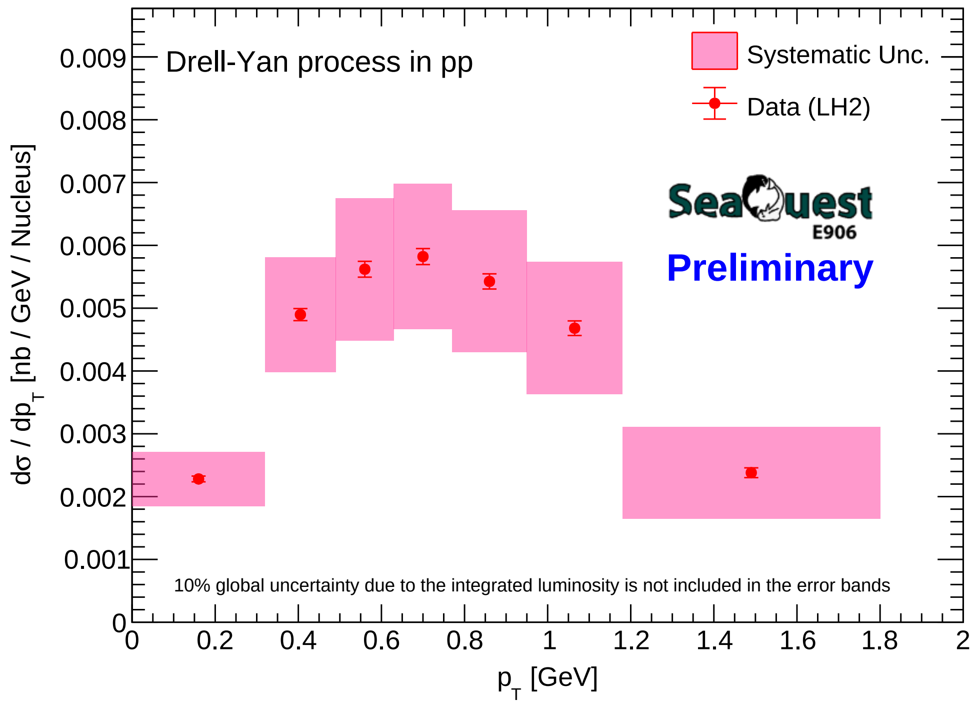

Figure 33: CrossSection_LH2_vs_pT_with_logo

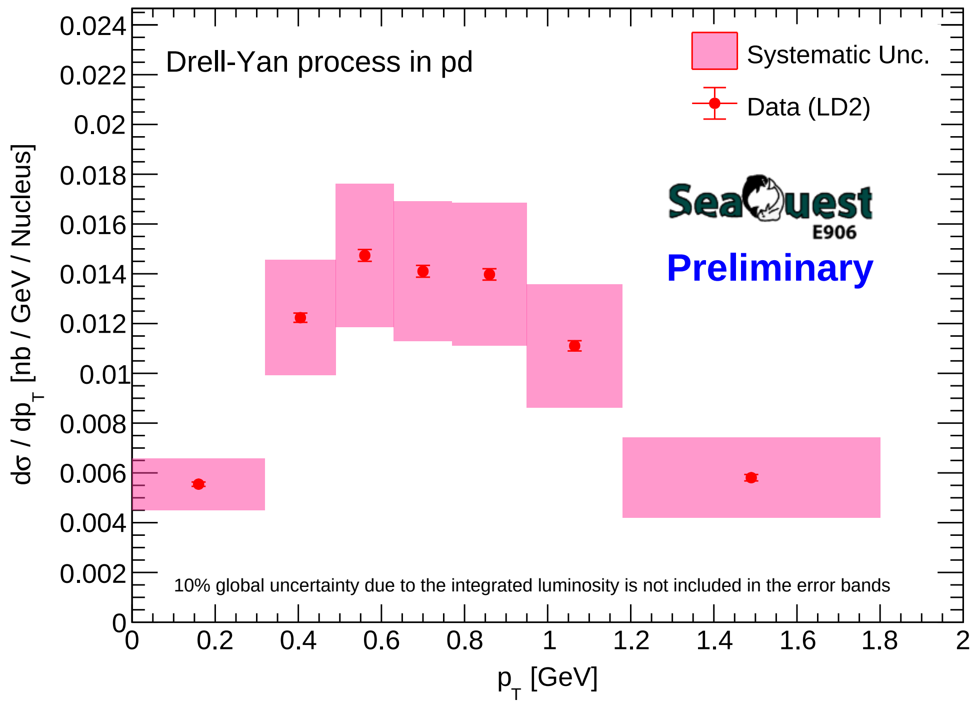

Figure 34: CrossSection_LD2_vs_pT_with_logo